

Juan David Muñoz-Bolaños

Prozessingenieur Messtechnologie

Rechenauerstraße 42, Innsbruck, Austria (Tirol)

+43 676 9115145

juan.munoz@i-med.ac.at | LinkedIn Profile



PROFESSIONAL SUMMARY

Physicist and Optical Engineer with extensive expertise in precision metrology, interferometry, and high-resolution instrument development. Proven track record in bringing complex optical systems from concept to industrial-grade prototypes. Strong foundation in Python/C++ automation, ZEMAX simulation, and quality assurance under industrial norms (ISO 10110/9001).

TECHNICAL COMPETENCIES

Optical Metrology & System Design

End-to-end design and assembly of multiphoton/confocal microscopes and interferometric setups. Expertise in wavefront sensing and diffraction-limited system alignment.

Simulation & Tolerancing

Advanced use of ZEMAX OpticStudio for systematic tolerance analysis, sensitivity characterization of custom objectives, and Zernike-based wave-field simulations in Python.

Software & Automation

Expert proficiency in C++, Python (NumPy, Pandas, Scikit-learn, JAX), and LabVIEW/FPGA for real-time hardware control, detector calibration, and automation of calibration routines.

Industrial Vision & QA

Hands-on experience with Halcon and Cognex VisionPro for production-line inspection. Proficient in MSA (Gage R&R) and designing custom illumination concepts (Bright-field/Dark-field).

Standards & Compliance

Strong awareness of ISO 10110 and ISO 9001 standards. Experienced in managing EH&S compliance and maintaining technical documentation for high-power laser environments.

PROFESSIONAL EXPERIENCE

Medical University of Innsbruck

PhD Researcher - Optical Metrology & Automation

Innsbruck, Austria

Jan 2022 – Present

- **Metrology System Build:** Designed and automated a custom wavefront sensing and phase retrieval system from scratch; wrote complete control stacks in C++/Python to interface cameras with deformable mirrors.
- **Optical Characterization:** Performed rigorous ZEMAX tolerance analysis to characterize aberration sensitivity in custom-built microscope objectives, validated via experimental interferometry.
- **R&D Project Leadership:** Led the hardware-software integration of a multiphoton microscope system, achieving diffraction-limited performance while managing vendor qualifications (Thorlabs, Edmund Optics).
- **Lab Operations:** Primary operator for a high-power laser lab, managing safety compliance (EH&S), equipment maintenance logs, and technical spec-sheet documentation.

Leibniz Institute of Photonic Technology (IPHT)

R&D Researcher (Photonics & Data)

Jena, Germany

Jun 2020 – Dec 2021

- **Spectrometer Prototyping:** Designed and built a 1064 nm fiber-probe Raman spectrometer for non-invasive clinical use; handled end-to-end optical layout, dispersive optic alignment, and detector calibration.
- **Data Science & ML:** Applied supervised Machine Learning to hyperspectral photonic data using custom Python packages, achieving high-accuracy tissue classification and accelerating experimental validation.

loops.

- **Interdisciplinary Research:** Collaborated on clinical spectroscopy projects, resulting in two peer-reviewed publications in Applied Sciences and ACS Photonics.

INTECOL S.A.S.

Junior Machine Vision System Engineer

Medellín, Colombia

Mar 2018 – Mar 2019

- **Industrial Vision Pipelines:** Engineered Halcon pipelines for fast-moving production lines (glass bottle inspection), implementing custom illumination concepts to solve liquid level and label printing quality issues.
- **System Commissioning:** Successfully deployed automated inspection solutions for clients like Renault, utilizing Cognex VisionPro for robotic part recognition and flame-surface treatment.
- **Process Reliability:** Conducted feasibility studies and Gage R&R assessments to ensure measurement traceability and process stability under industrial conditions.

EDUCATION

Medical University of Innsbruck

Ph.D. in Adaptive Optics & Microscopy

Austria

Jan 2022 – Present

Friedrich Schiller University Jena

M.Sc. in Photonics

Germany

2019 – 2021

University of Cauca

B.Sc. in Physics Engineering

Colombia

2012 – 2018

PUBLICATIONS & AWARDS

- **Awards:** Selected for ASML and ZEISS Summer Schools; awarded COLFUTURO Grant 2026 for top Colombian professionals.
- Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. ACS Photonics 12(1), 176–184 (2025).
- Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer. Applied Sciences, 14(11), 4726 (2024).
- Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. Biomedical Optics Express 14(4), 1562–1578 (2023).

LANGUAGES

English (Fluent / Professional),
German (B2 - Working Proficiency),
Spanish (Native)

REFERENCES

Prof. Monika Ritsch-Marte

Head of Institute for Biomedical Physics
monika.ritsch-marte@i-med.ac.at

Medical University of Innsbruck

Dr. Christoph Krafft

Group leader: Spectroscopy Imaging
christoph.krafft@leibniz-ipht.de

Leibniz Institute of Photonic Technology